

BOOTCAMP MLOps



BOOTCAMP MLOps:

What we will learn to day:
Models and Services monitoring

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Goals

- Understand the importance of models and services monitoring.
- Know how to setup and log your metrics on services.
- Hands-on lab with Grafana and EvidentlyAI

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Introduction to Monitoring

Drift Detection 101

Introduction to service logging and monitoring

Introduction to Evidently AI

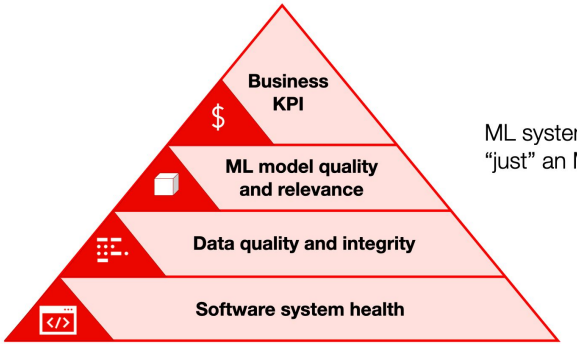
Hands-on lab

Introduction to Monitoring

Why Monitoring Is Critical in ML

Machine learning models degrade over time due to real-world changes.
Monitoring ensures:

- Reliability of predictions
- Prevention of silent model failure
- Compliance with organizational or regulatory standards
- Continuous operational stability



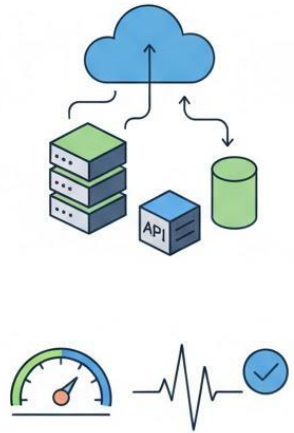
ML system is more than "just" an ML model

Two Complementary Monitoring Domains

Modern ML systems require two separate but complementary monitoring approaches:

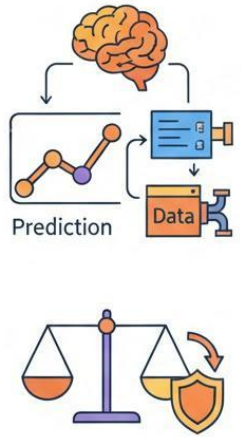
- **Service Monitoring** – Ensures the API and infrastructure are functioning properly.
- **ML Monitoring** – Ensures the data and predictions remain valid and consistent with training conditions.

Service Monitoring



Ensures the API and are functionng properly.

ML Monitoring



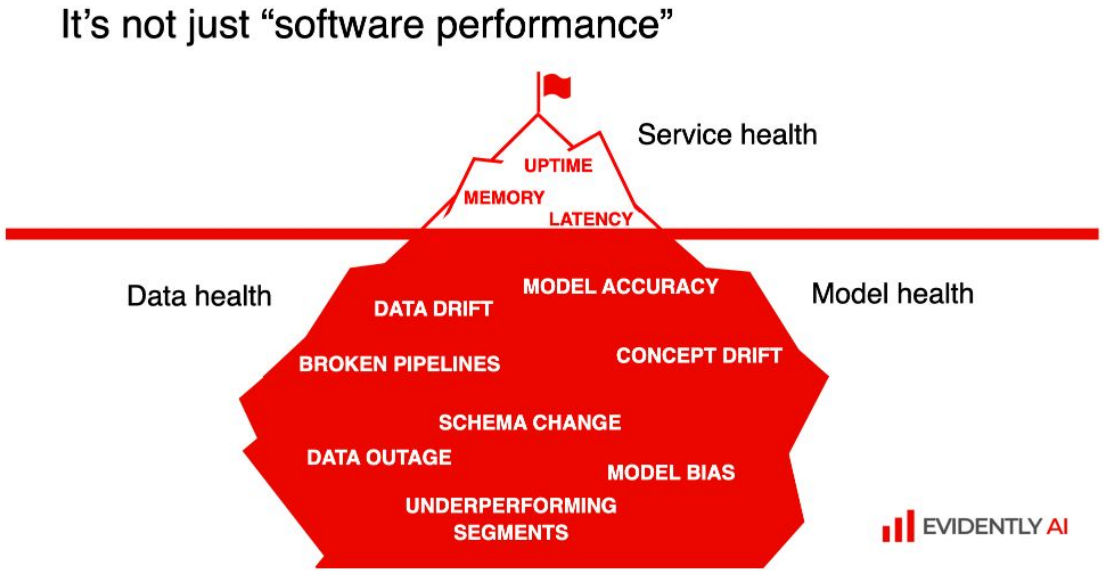
Ensures the data and predictions remain valid and consistent with training conditions

Hierarchy of MLOps Monitoring Needs

Monitoring can be organized into three layers:

- **Service Health:** Infrastructure and API level.
- **Data Health:** Input features and distribution integrity.
- **Model Health:** Prediction accuracy and concept stability.

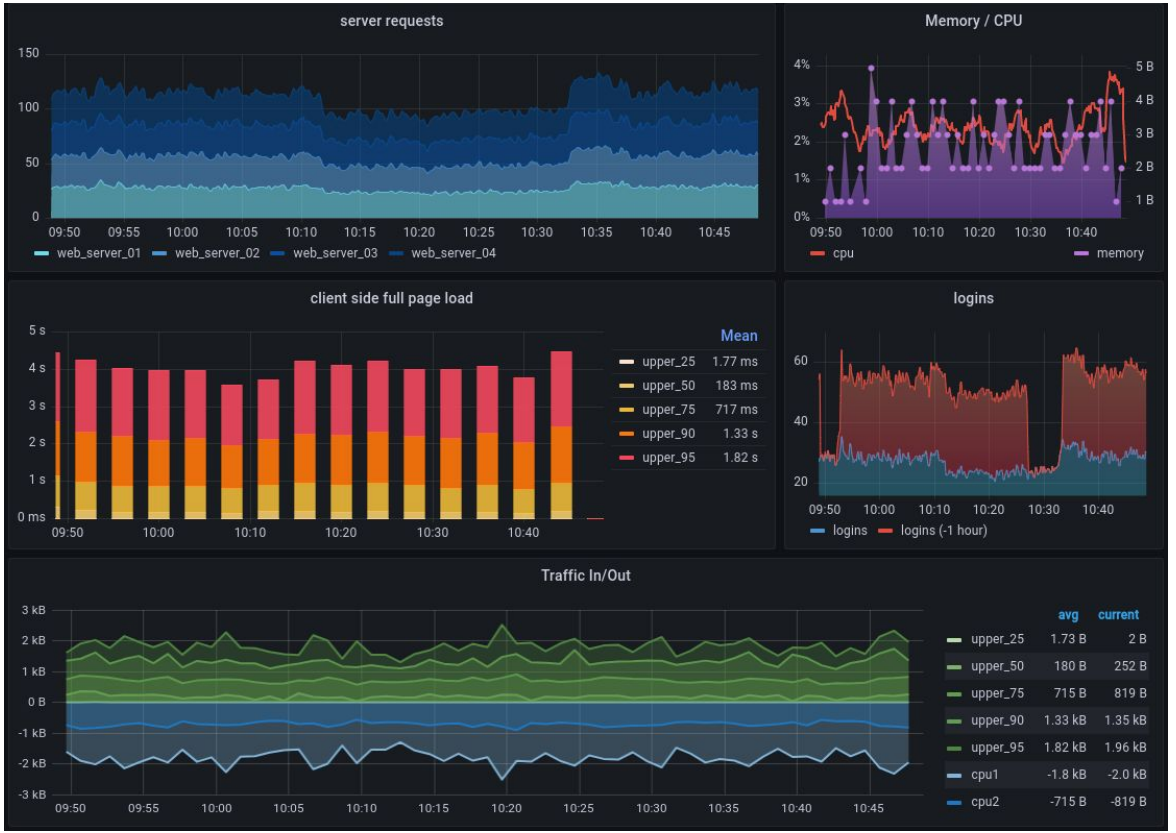
Each upper layer depends on the stability of the layers below.



Layer 1: Service Health

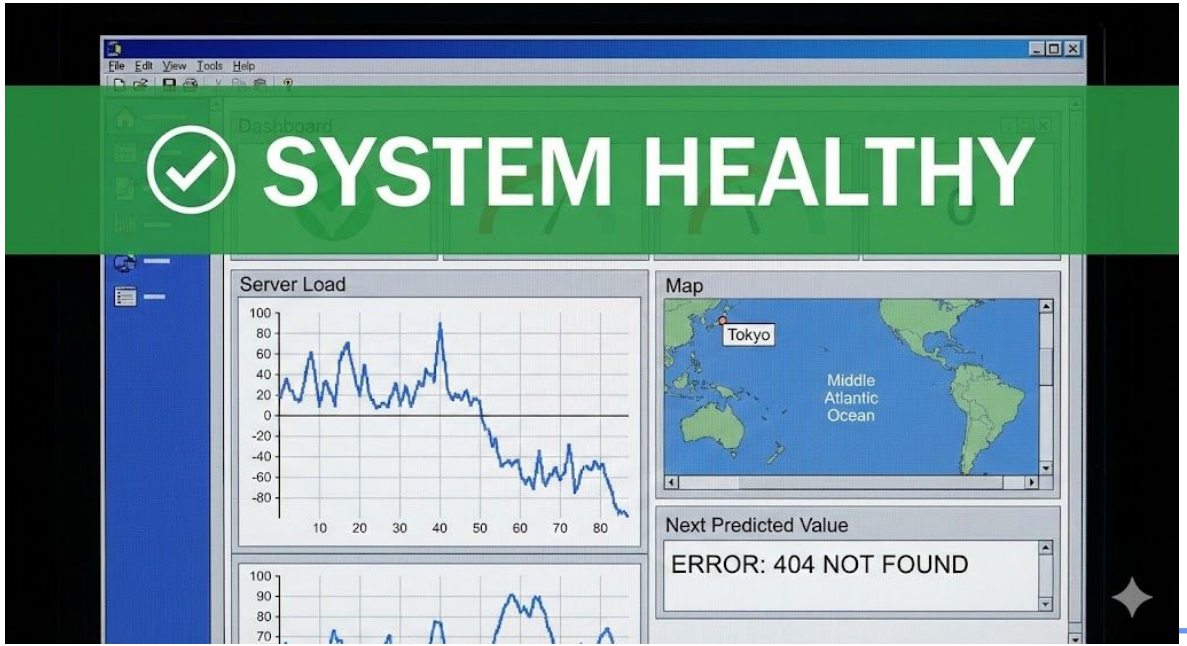
Ensures the ML service is reachable and functional. Key checks include:

- API response latency
- Request throughput
- Error rates (4xx, 5xx)
- CPU, memory, and container restarts
- Uptime SLAs



Limitations of Service Health Metrics

A perfectly functioning API does not guarantee correct predictions.
Example:
A recommendation API can respond fast with a 200 OK, while serving outdated or biased results due to data drift.

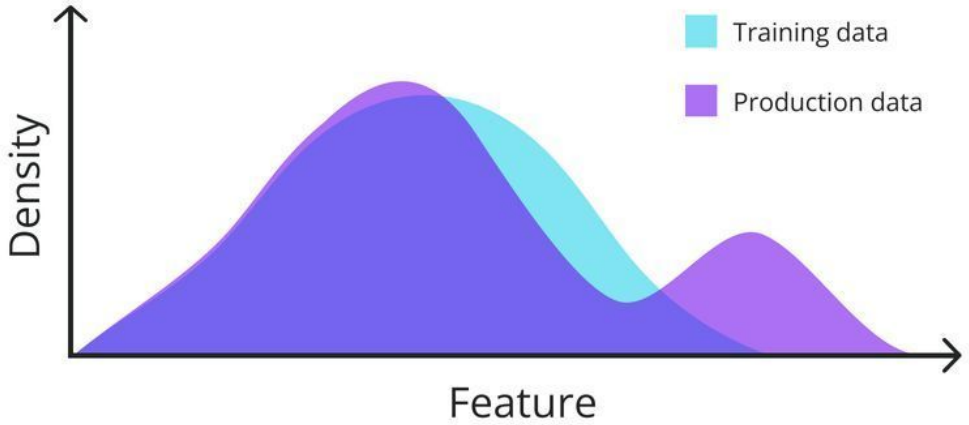


Layer 2: Data Health

Examines the integrity and stability of incoming data.

Focus areas:

- Feature distribution
- Missing or unexpected values
- Schema consistency
- Outliers and anomalies
- Shifts compared to training data

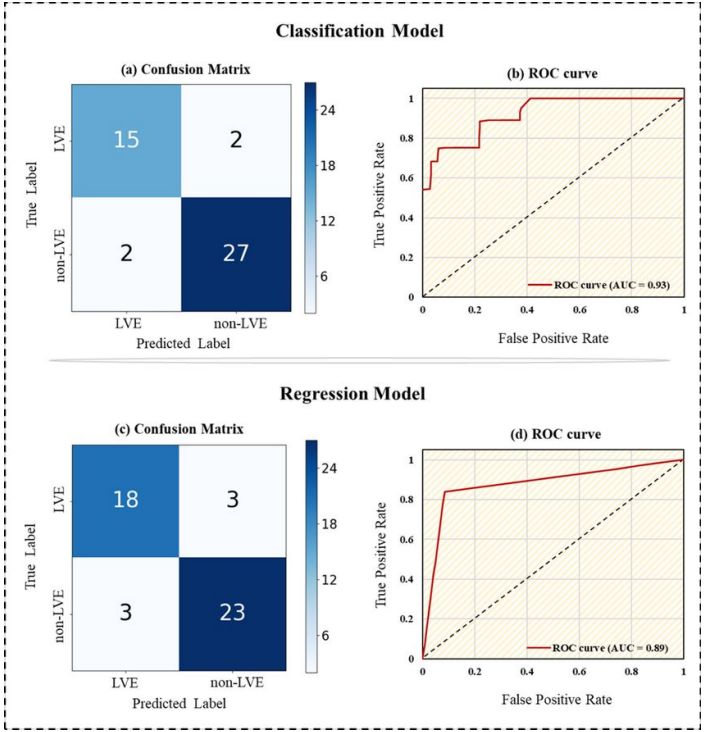


Layer 3: Model Health

Tracks whether the model is still predicting correctly.
Requires ground truth labels, which often arrive with delay.

Key indicators:

- Accuracy, precision, recall
- Prediction drift
- Confidence score patterns
- Calibration errors

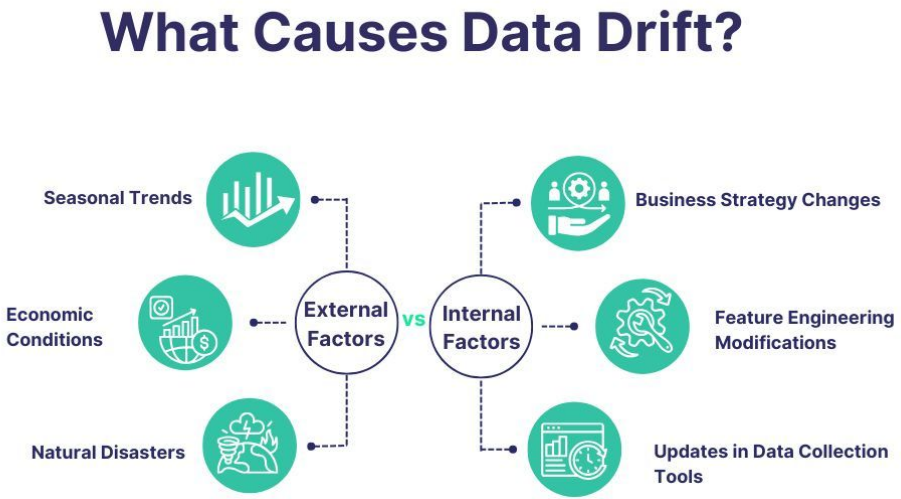


Drift Detection 101

Root Causes of Drift

Possible sources:

- Market or environmental changes
- Seasonality effects
- Data pipeline modifications
- Product changes (e.g., UI updates)
- Changing user behaviors
- Sensor degradation (IoT)



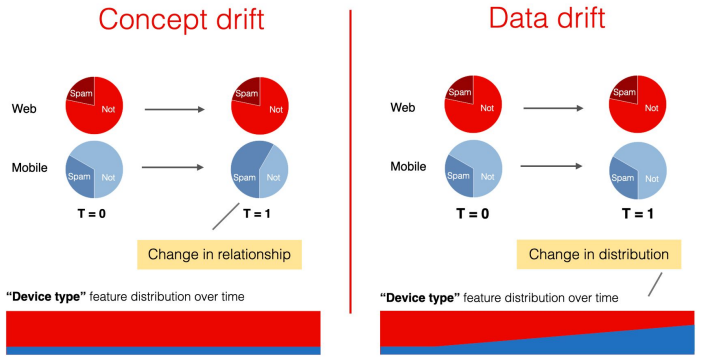
Three Types of Drift

Data Drift: The input features' distribution changes.

Concept Drift: The relationship between features and labels changes.

Prediction Drift: The prediction outputs shift abnormally.

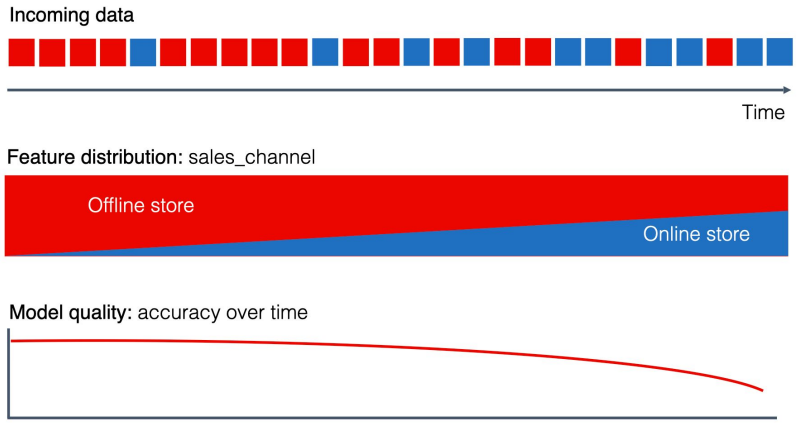
Each drift type requires different detection and remediation strategies.



https://www.evidentlyai.com/ml-in-production/concept-drift

What Is Data Drift?

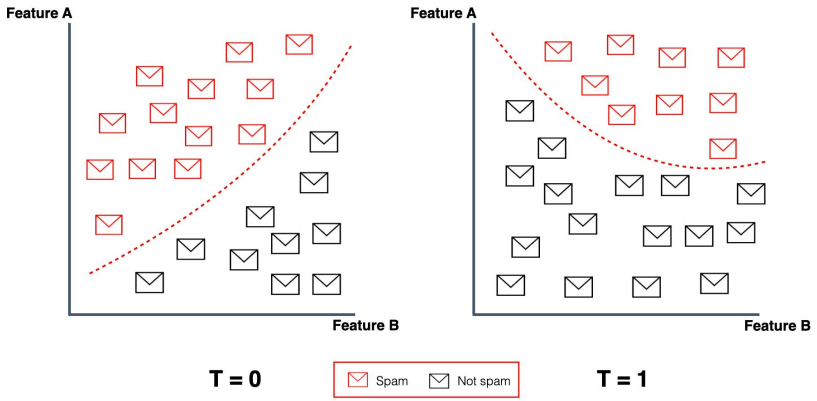
Data drift occurs when the distribution of input features in production differs significantly from the reference dataset (usually training or validation data).
It's often the first sign of model decay.



https://www.evidentlyai.com/ml-in-production/data-drift

What Is Concept Drifts

Concept drift refers to changes in the data patterns and relationships that the ML model has learned, potentially causing a decline in the production model quality.



https://www.evidentlyai.com/ml-in-production/data-drift

The Impact of Untreated Drift

Consequences include:

- Lower model performance
- Incorrect or unfair decisions
- Lost revenue
- Regulatory issues
- Reputational damage
- Monitoring allows early detection to avoid business impact.



https://www.evidentlyai.com/ml-in-production/concept-drift

Statistical Tests for Drift Detection

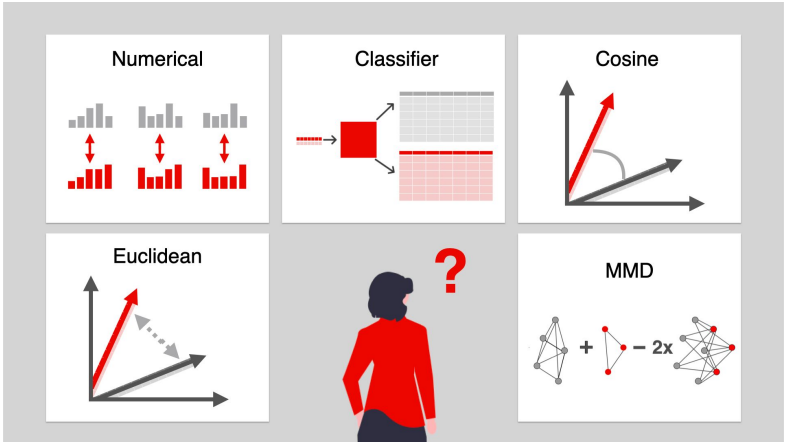
Common statistical methods include:

Kolmogorov–Smirnov Test (numerical features)

Chi-Square Test (categorical features)

Jensen–Shannon Divergence

Population Stability Index (PSI)



https://www.evidentlyai.com/ml-in-production/concept-drift

Drift Metrics and Interpretation

Metrics help determine the severity of drift:

- Drift Score
- p-values
- Feature-level drift rankings
- Threshold-based classification (drift/no drift)

Useful for deciding if retraining is necessary.

	PSI	KL	JS	KS	Embedding Drift
Support Numeric Features	✓ Yes	✓ Yes	✓ Yes	✓ Yes	Not Natively
Support Categorical Features	✓ Yes	✓ Yes	✓ Yes	✓ Yes	Not Natively
Support Image	✗ No	✗ No	✗ No	✗ No	✓ Yes
Support Text	✗ No	✗ No	✗ No	✗ No	✓ Yes
Symmetric	✓ Yes	✗ No	✓ Yes	✓ Yes	✓ Yes
Handles Empty Bins	Not Natively (Yes with ODB)	Not Natively (Yes with ODB)	✓ Yes	✓ Yes	N/A
Threshold	*In Practice Needs Auto*				
Multi-Variate Drift	✗ No	✗ No	✗ No	✗ No	✓ Yes
Drift of Clusters	✗ No	✗ No	✗ No	✗ No	✓ Yes

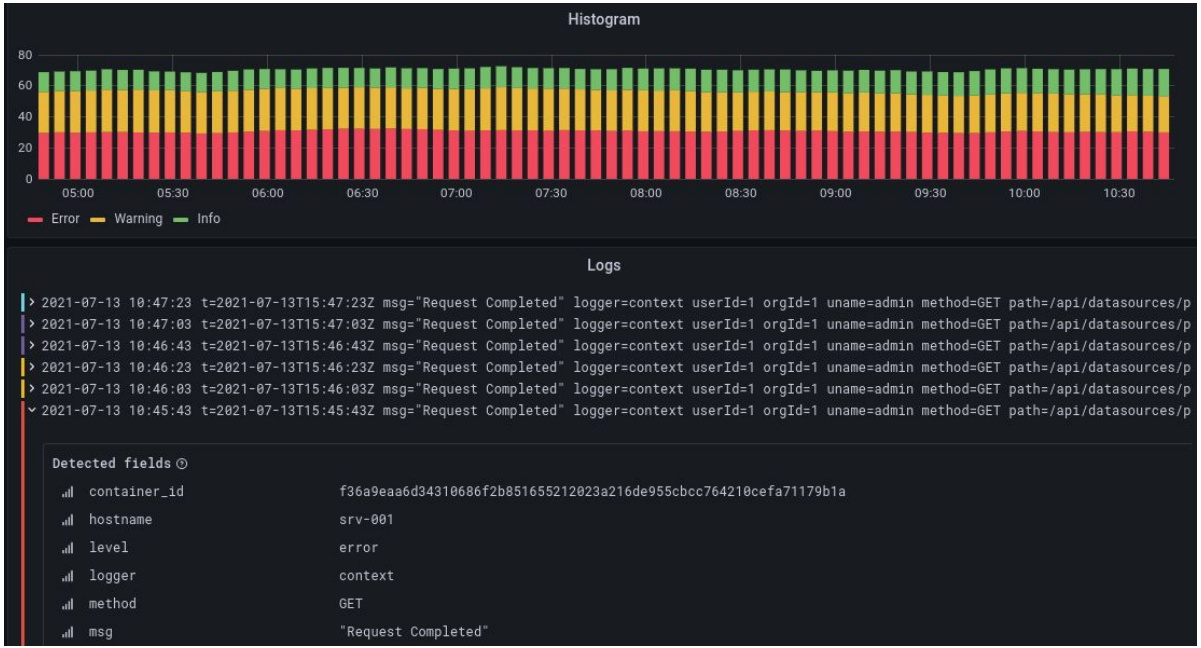
https://www.evidentlyai.com/ml-in-production/concept-drift

Introduction to service logging and monitoring

Why Logging Is Foundational

Reliable monitoring requires a complete, structured trail of production data:

- Captures real inference inputs
- Supports debugging and audit
- Enables drift and performance analysis
- Enables reproducibility of predictions



https://www.evidentlyai.com/ml-in-production/concept-drift

What Should Be Logged?

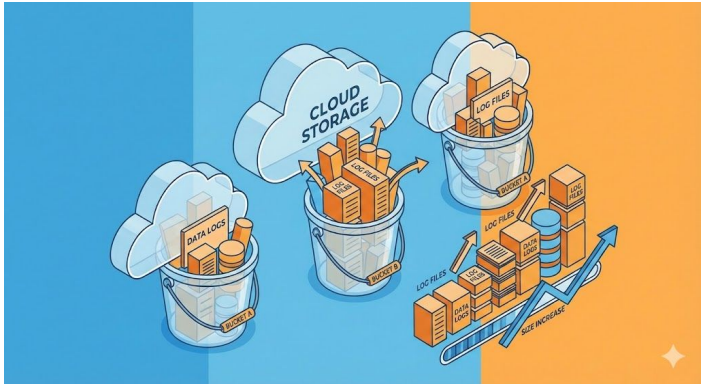
Recommended fields:

- Raw input features
- Preprocessed features (if different)
- Model predictions
- Confidence scores
- Timestamp
- Metadata (client, version, trace ID)

Logging Challenges in Production

Key challenges include:

- High data volumes
- PII handling and compliance
- Evolving data schemas
- Choosing the correct storage strategy
- Managing scale and retention



https://www.evidentlyai.com/ml-in-production/concept-drift

Tools for Service Monitoring

Widely used tools for infrastructure and API health:

Prometheus (metrics collection)

Grafana (visual dashboards)

OpenTelemetry

Datadog, **New Relic**

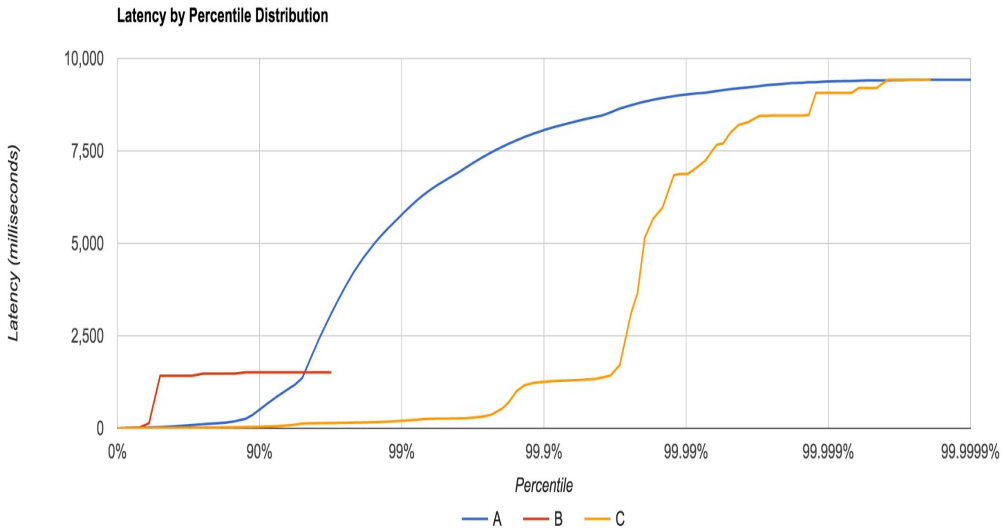
These tools excel at operational metrics.



Example Service Metrics

Common metrics monitored:

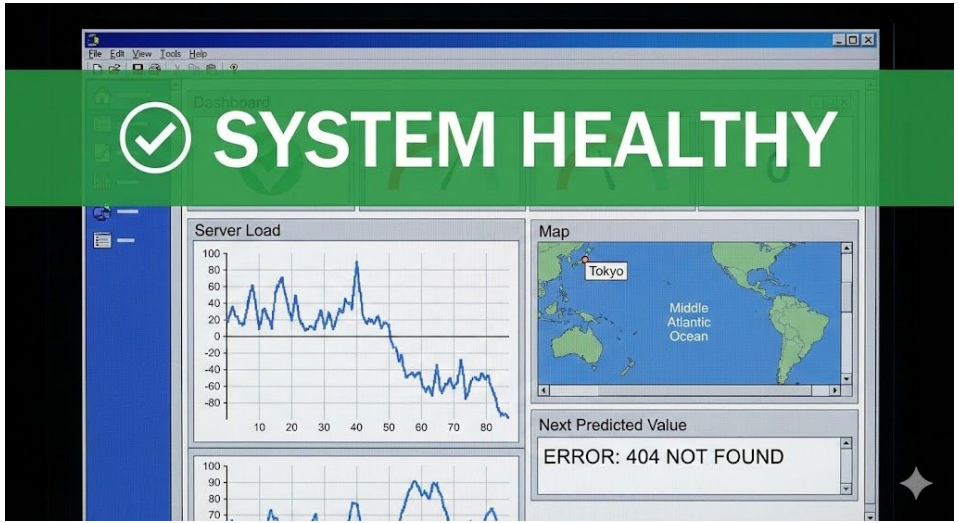
- Request latency (p95, p99)
- Request throughput
- Error rate (4xx/5xx)
- Uptime and restarts
- CPU, RAM, disk usage



https://www.evidentlyai.com/ml-in-production/concept-drift

Why Service Metrics Can Miss ML Failures

Service metrics focus on system behavior, not **prediction quality**.
A healthy API can still produce harmful predictions due to drift or poor data quality.



https://www.evidentlyai.com/ml-in-production/concept-drift

Introduction to Evidently AI

Tools for ML Monitoring

Specialized tools track data drift, predictions, and model behavior:

- Evidently AI
- Arize
- WhyLabs
- Fiddler AI

These tools focus on ML-specific signals unavailable in service-only monitoring.









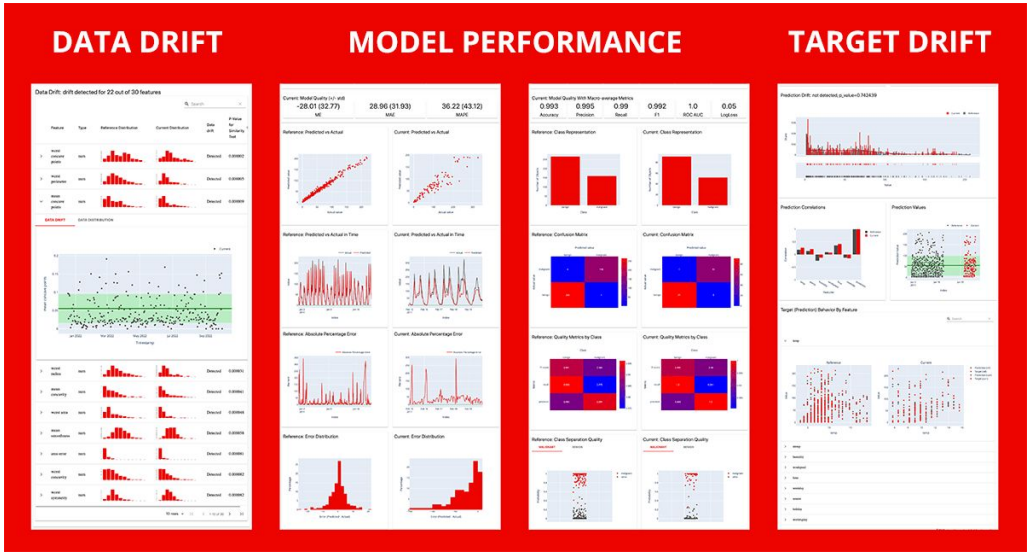
https://www.evidentlyai.com/ml-in-production/concept-drift

Introduction to Evidently AI

Open-source library designed for ML observability, offering:

- Data Drift reports
- Target Drift reports
- Data Quality checks
- HTML dashboards
- Feature-level analysis

Widely used due to simplicity and rich visualizations.



https://www.evidentlyai.com/ml-in-production/concept-drift

Structure of an Evidently Report

A typical report contains:

- Summary metrics
- Feature-by-feature drift statistics
- Distribution comparisons
- Heatmaps
- Alerts when thresholds are exceeded



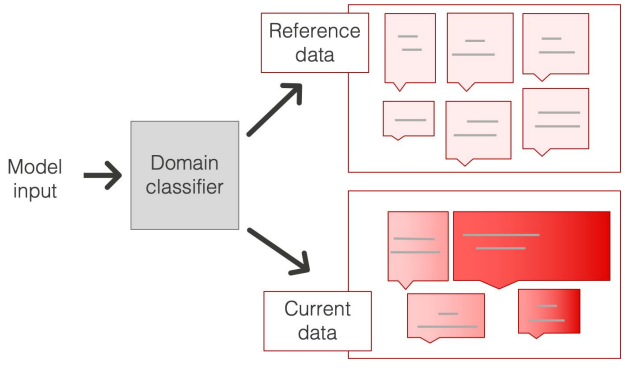
https://www.evidentlyai.com/ml-in-production/concept-drift

Reference vs. Production Data

The core concept of drift detection:

Reference Dataset: Stable dataset (training/validation).

Production Dataset: Recent data from live traffic. Comparing these two reveals drift patterns.

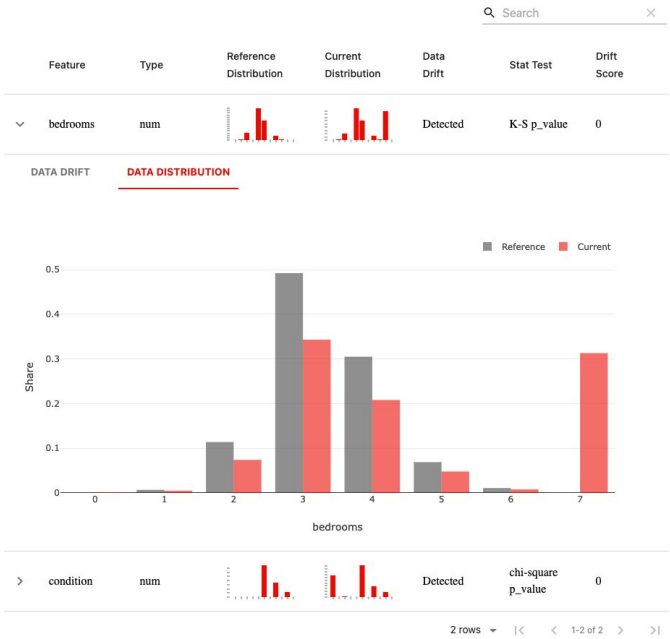


Visualizing Drift in Data

Visual plots help explain drift clearly:

- Density plots
- Categorical frequency charts
- Box plots
- Scatter plots

These make drift interpretable for engineers and business stakeholders.



https://www.evidentlyai.com/ml-in-production/concept-drift

The Importance of Visual Reports

Visualization allows fast diagnosis:

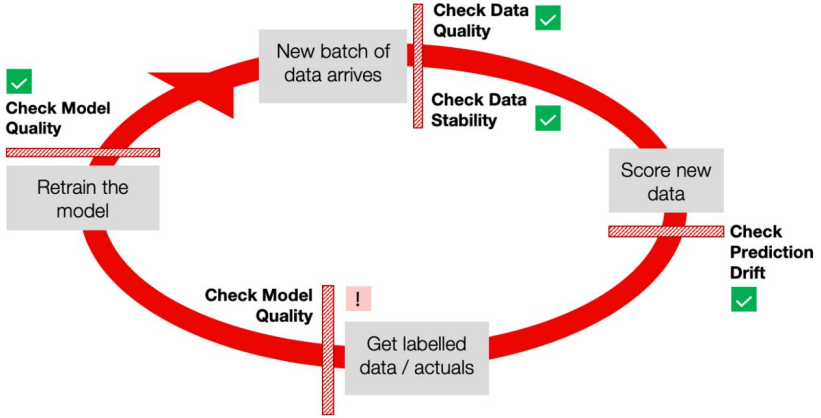
- Understand which features changed
- Spot anomalies
- Explain issues to non-technical audiences
- Review the magnitude and nature of drift

https://www.evidentlyai.com/ml-in-production/concept-drift

The ML Monitoring Feedback Loop

Monitoring completes a full production feedback cycle:

- Capture live data
- Analyze drift and quality
- Issue alerts
- Decide on mitigation steps
- Retrain or adjust model
- Deploy and repeat



https://www.evidentlyai.com/ml-in-production/concept-drift

Keys takeaway

Service health ≠ model health

Data drift is inevitable and must be detected

Logging is the foundation of ML observability

Evidently AI provides clear drift reports and visual insights

Monitoring is essential to sustaining trustworthy ML systems

COLE

https://www.evidentlyai.com/ml-in-production/concept-drift

Your mission

1. Environment Setup
2. FastAPI + Loguru + Prometheus
3. Prometheus + Grafana
4. Simulate Traffic + Drift
5. Evidently AI Drift Report

Q&A



Thank you for listening!

